PCB Inspection Report

Timestamp	2026-04-04 16:31:40
Status	DEFECTIVE
Risk Level	HIGH
Failure Probability	70.06%

Detected Defects

#	Label	Confidence	BBox
1	Missing_hole	0.7954	[1192, 280, 1214, 301]
2	Missing_hole	0.6899	[802, 55, 848, 100]
3	Missing_hole	0.6693	[705, 55, 745, 98]
4	Missing_hole	0.649	[532, 274, 553, 297]
5	Missing_hole	0.5945	[108, 198, 131, 222]
6	Missing_hole	0.3384	[598, 56, 639, 98]
7	Missing_hole	0.3038	[954, 704, 973, 722]

Explanation

Root Cause Analysis
The detection of seven distinct Missing_hole defects, with confidence levels ranging from 0.7954 to 0.3038, points to a systemic failure within the PCB fabrication process, specifically at the drilling or plating stages. The higher confidence values confirm unambiguous missing features, while lower confidence detections may indicate partially drilled holes or holes obstructed by plating residue. Likely manufacturing process contributors include worn or broken drill bits, incorrect NC drill programming, inadequate drill machine maintenance leading to positional inaccuracies or insufficient drill depth, or issues during the electroless/electrolytic plating process causing complete hole closure. The distributed nature of the bounding boxes suggests a general process control issue rather than an isolated tool-specific failure.
Operational Risks
This PCB presents a HIGH operational risk due to the numerous Missing_hole defects, directly contributing to the calculated 70.06% failure probability. If these missing holes are intended for critical through-hole component leads, the absence of connection points will prevent component population and result in immediate open circuits, rendering the board non-functional. If they are intended as vias, essential electrical signal paths or power/ground connections will be interrupted, leading to complete circuit failure. Furthermore, if these holes are for mechanical mounting or alignment, the board will not integrate correctly into its final assembly, potentially causing misalignment, physical stress on other components, or complete assembly line stoppage. The number and confidence of the defects indicate a severe compromise to the board's designed functionality.
Failure Prediction
Given the type and prevalence of the Missing_hole defects, this PCB is highly predicted to fail during initial electrical test or functional verification. The absence of critical electrical pathways, confirmed by high confidence detections such as the missing hole at bbox=[1192, 280, 1214, 301]

with 0.7954 confidence, will prevent proper circuit operation immediately upon power application. Even the lower confidence detections, such as those at bbox=[598, 56, 639, 98] and bbox=[954, 704, 973, 722], while potentially less severe, collectively contribute to a compromised board integrity that significantly elevates the cumulative failure probability to 70.06%. Latent failures are also a concern if partially formed holes lead to intermittent connections or reduced reliability over operational lifespan.

Repair Explanation

For defects of type Missing_hole, especially those identified with high confidence, the PCB is generally considered non-repairable. Re-drilling holes into a fabricated board, particularly if they are intended to be plated through-holes, poses significant risks including damage to internal traces, delamination of layers, creation of burrs that could short adjacent layers, and compromised plating integrity. Such interventions would severely degrade the board's reliability and often exceed the cost of manufacturing a new board. Attempting to manually create these holes would introduce unpredictable variations and negate any quality or reliability guarantees. Therefore, this defective PCB should be scrapped.

Engineering Insights

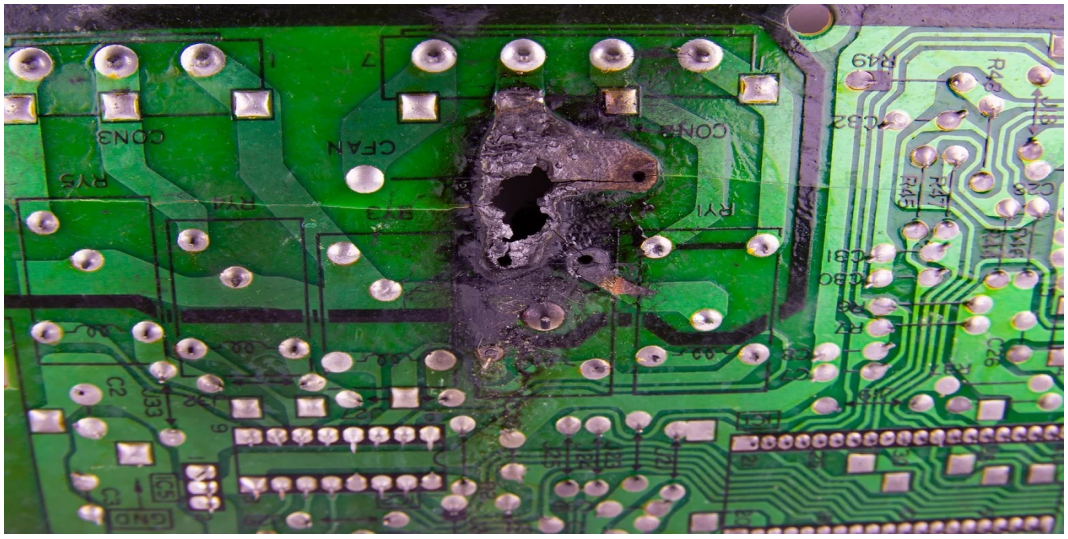
The consistent detection of Missing_hole defects across multiple locations necessitates immediate investigation into the PCB fabrication facility's drilling and plating processes. Engineering teams should review the NC drill files against the original design data to rule out programming errors. Emphasis must be placed on drill bit management, including wear monitoring, replacement schedules, and ensuring correct drill bit selection for specified hole sizes. Calibration and maintenance of drilling machines are critical to ensure accurate positional registration and consistent drill depth. Implementation of in-process optical inspection immediately after the drilling stage, prior to plating, would allow for earlier detection of missing or partially drilled holes. For plated holes, cross-sectional analysis or micro-resistivity measurements could further diagnose plating issues. The spatial distribution of the defects from the bounding boxes should be analyzed for patterns that might indicate specific machine head issues, panelization problems, or material stress points during drilling.

Repair Suggestions

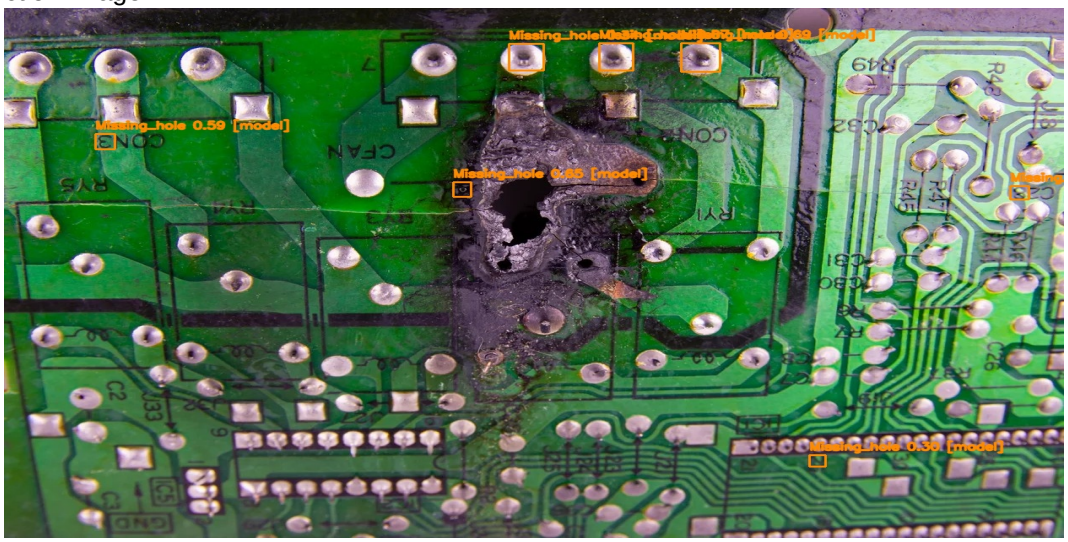
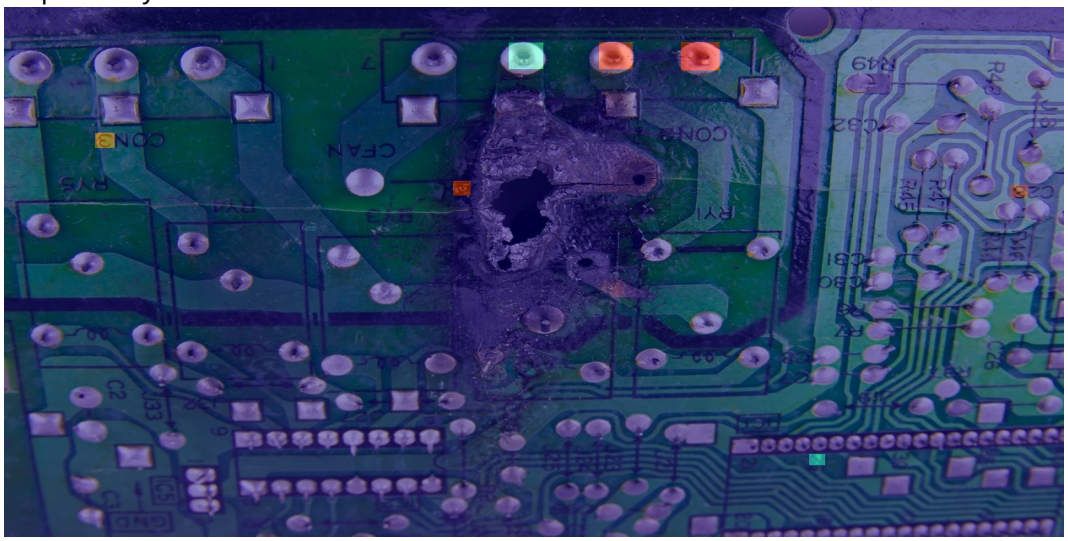
- Missing_hole: Replace the missing component and verify orientation and BOM value before reflow.

Inspection Images

Original Image



Detection Image



Heatmap Overlay